



Lab Coat Guidelines

1. Introduction

Making sure that the University is a safe and healthy workplace is the right and responsibility of everyone working at the University. Furthermore, the Ontario Occupational Health and Safety Act requires employers and supervisors to ‘take every reasonable precaution in the circumstances for the protection of a worker’. The Act requires supervisors to ensure that a worker ‘uses or wears the equipment, protective devices or clothing that the worker’s employer requires to be used or worn.’

Lab coats are protective equipment that must be worn whenever and wherever hazardous materials are being used in the laboratory including chemicals, biological materials, and radioactive materials. Laboratory coats are intended to protect against minor splashes or spills, and to minimize contamination of street clothing with materials used in the laboratory. Laboratory coats must be made of material suitable for the work environment, the materials handled, and the tasks performed. Lab coats are to be worn in combination with full-length pants (or legs coverings with equivalent protection), socks, and closed-toe shoes to prevent exposed skin in the arms and legs.

2. Application

This document applies to any University of Toronto employees, students, Postdoctoral Fellows, visitors, and volunteers that enter a University of Toronto wet lab, where their activities in the lab put them at risk of chemical, biological or radiological contamination.

3. Definitions

Lab User: Any individual who is assigned to a lab to conduct laboratory activities, or to supervise other lab users.

Lab Visitor: Any individual in the lab who is not a Lab User.

Personal Protective Equipment (PPE): Equipment or clothing that is intended to protect the wearer from specific hazards.

Wet Lab: A Wet Lab is any lab where chemical, biological or radioactive materials are used.

4. Responsibilities

4.1 Directors, Department Heads

It is the responsibility of Directors and Department Heads to ensure employees working in a wet lab are aware of the guidelines for lab coats.



4.2 Principal Investigators (PI) and Lab Managers

Principal Investigators and Lab Managers must determine applicable personal protective equipment (PPE) requirements based on health and safety hazards in the laboratory. The University of Toronto Office of Environmental Health and Safety (EHS) can assist with evaluating hazards and PPE requirements.

Any University of Toronto PI that supervises wet lab space is responsible for:

- a) ensuring that any employees or students they supervise receive appropriate EHS laboratory safety training which includes a component on PPE, in addition to work-specific instructions and procedures related to their lab environment;
- b) conducting local risk assessments to determine when and where lab users and lab visitors are at risk of physical contact exposure, and the protective equipment that would be required;
- c) advising lab users of the existence of any potential or actual danger to health or safety;
- d) providing any required PPE; and
- e) ensuring that a lab user works in the manner and with required PPE, measures and procedures required.

4.3 Employees and Student Lab Users or Lab Visitors

It is the responsibility of any University of Toronto lab users or lab visitors to adhere to the procedural requirements specified within their lab environment. In addition, lab users shall:

- a) Use or wear lab coats and any protective devices or clothing required to be used or worn;
- b) Report to their supervisor any knowledge of the absence of, or defects in, any equipment or protective device (e.g. guarding, safety switch etc.) which may endanger them or other lab occupants.

Guest lab coats should be made available in the laboratory when necessary. It is recommended that laboratories keep additional lab coats on hand for users in the event of a spill or when a spare lab coat is required.

5. Lab Coat Guidelines

A lab coat must:

- a) Be appropriate for the materials in use and activities in labs. Select the coat/gown based on hazards used in the laboratory. For example, a fire-resistant coat is recommended for work with pyrophoric or flammable material;
- b) Fit properly; should be knee-length or slightly below;
- c) Be properly fastened, ideally with snaps;
- d) Have sleeves that fully cover the arms to the wrist (short sleeved lab coats are not permitted for wet lab use);
- e) Be removed before leaving the wet lab. Under specific circumstances, personnel may develop and implement an in-house SOP which outlines exceptions to this rule (e.g., where there is a need to carry materials through a hallway to another wet lab). In these circumstances, the lab coat worn must be devoid of biological, chemical and/or radiological contaminants).



Notes

- If there is a hazard from lab coat sleeves becoming entangled/catching on equipment, lab coats with knitted/elastic cuffs may be purchased and should be in accordance with specific machine guarding safety.
- A lab coat should never be rolled up to expose any portions of the hand or arm that is not already protected by an appropriate glove.

5.1 Choosing the right lab coat

Lab coats are offered in a variety of materials and not all materials are suitable for all activities. The choice of lab coat is laboratory environment specific and could also be experiment specific such as when using pyrophoric materials.

Material	Splash Resistance/ Chemical Resistance	Flame Resistance	Uses/ Comments
Synthetic/Cotton Blends Recommended 65/35 and 40/60 for chemical research lab setting	Splash Resistant Unknown chemical resistance	Most combustible Not considered appropriate for working with flammables and should never be used while working with pyrophoric materials.	100% Polyester, 80/20, 65/35 and 40/60 polyester/cotton blend lab coats are the most common lab coats. Good for clinical settings (hospitals, clinical labs) and labs handling biological materials. 100% polyester and 80/20 blend lab coats are NOT recommended for chemical laboratories.
100% Cotton	Less resistant than blends and are degraded by acids.	Superior to synthetic blends for fire-resistance, and are a good affordable compromise for chemical safety than the more expensive Fire-Resistant lab coats.	Comfortable Standard undergraduate lab coat
Flame Resistant treated lab coats	Not generally tested for chemical resistance.	Generally, will not lose flame resistance with laundering over typical use life, specific manufacturer recommendations must be followed.	Better for labs with significant fire hazard, with an understanding of the limitations of the testing criteria for flame resistance. More costly



Material	Splash Resistance/ Chemical Resistance	Flame Resistance	Uses/ Comments
Dupont Nomex®	Unknown splash and chemical resistance	Yes	Expensive Good for lab environments where there is a risk of arc flash or flash fire and recommended for working with pyrophoric materials.

5.2 Lab Coat Cleaning and Disposal

5.2.1 General Lab Coat Cleaning

Non-disposable lab coats should be washed on a regular basis with washing intervals not normally exceeding once a month to ensure they are clean for lab users to wear. Lab coats should be washed according to lab activities. Nomex® and Fire-Resistant lab coats are not compatible with bleach and need to be washed with care to avoid damaging the fire-retardant properties.

It is recommended to:

- 1) Have a dedicated on-site facility; or
- 2) Have off-site washing services performed by a contractor with whom a written agreement is established. The contract should clearly outline the contractor's acknowledgment of the possible presence of hazardous materials.

Note: If no washing option is available and if allowed by a local risk assessment, lab coats should be double bagged for home transportation and laundered separately.

5.2.2 Chemical: Cleaning and Disposal

Contaminated lab coats should be discarded following any applicable U of T hazardous waste disposal procedures and not washed if contaminated with:

- 1) Strong acids or concentrated corrosives;
- 2) Materials that pass through Nitrile gloves (e.g. organometallics like methyl mercury); or
- 3) Greater than 25 mL of:
 - a) Carcinogens;
 - b) Teratogens;
 - c) Toxic materials with an LD50<50mg/kg.

If the spill does not meet any of the above conditions a coat may be washed and reused.

Lab coats contaminated by hazardous material spills must be promptly bagged and clearly labeled as hazardous waste for proper disposal.

In addition, any lab coat with significant holes, tears or other physical damage should be replaced.



5.2.3 Biological: Cleaning and Disposal

Where a known or suspect contamination/spill from ANY biological agent occurs (regardless of Risk Group of the biological agent), any contaminated clothing and the lab coat must be decontaminated with an effective disinfectant for an appropriate contact time (e.g., freshly prepared 1% sodium hypochlorite for 30 minutes) and then rinsed in water prior to laundering.

Lab coats should not be decontaminated with disinfectants if they are additionally contaminated with:

- a) Chemicals (including bleach); or
- b) Radioactive material.

Contact your Health and Safety Officer for further guidance.

5.2.4 Radiation: Cleaning and Disposal

If a spill on the lab coat involves any amount of a radioactive material:

- 1) The Designated Radiation Safety Officer or the Manager, Research Regulatory Affairs of the Radiation Protection Services (RPS) must be informed immediately.
- 2) The lab coat must be placed in a sealed plastic bag and provided directly to the Designated Radiation Safety Officer at the earliest opportunity.

6. Lab Coat Availability

Lab coats are available in a variety of fabrics and blends. Lab users should choose the appropriate one according to laboratory activities and in consultation with the laboratory PI/supervisor, or departmental policy. Lab coats are available at:

- 1) The University of Toronto Bookstores at the St. George, Scarborough and Mississauga campuses;
- 2) The University of Toronto Chemical Stores;
- 3) The University of Toronto Medical Stores, and;
- 4) Other laboratory suppliers.